

PRACTICAL WORK 3

Data Visualization: Multidimensional data

Dataset (see attached files)

- Wine quality dataset
 - <https://www.kaggle.com/datasets/yasserh/wine-quality-dataset>
- Diabetes dataset
 - <https://www.kaggle.com/datasets/akshaydattatraykhare/diabetes-dataset>

Task

To explore and understand different techniques for visualizing univariate, bivariate, multivariate, and high-dimensional data.

- Univariate Analysis
- Discrete, categorical attribute in 1-D
- Multivariate Analysis
- Parallel coordinates
- Two Continuous Numeric attributes
- Joint Plot for alcohol and quality
- Using boxplot and Violin plot: Compare distributions of alcohol content across quality levels
- Visualizing three dimensions: using Use 3D scatter plots for alcohol, density, pH.
- Visualize distribution of alcohol across different quality levels.

Deliverables

- Write a summarized report describing each visualization results.
 - Include the screenshots of the visualization plots
- Submit the code